

Pharmacology of Analgesics II: Drug used in Neuropathic Pain Centrally Acting Muscle Relaxants



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- ☀ อธิบายกลไกการออกฤทธิ์ของยาที่ใช้ในการรักษา **neuropathic pain** ได้
- ☀ อธิบายอาการไม่พึงประสงค์ ข้อควรระวัง ข้อห้ามใช้ และ อันตรกิริยาระหว่างยาที่สำคัญของยาที่ใช้ในการรักษา **neuropathic pain** ได้
- ☀ ยกตัวอย่างข้อบ่งใช้เฉพาะของยาที่ใช้ในการรักษา **neuropathic pain** ได้

- ☀ จัดแบ่งยาคลายกล้ามเนื้อตามกลไกการออกฤทธิ์ได้
- ☀ อธิบายอาการไม่พึงประสงค์ ข้อควรระวัง ข้อห้ามใช้และ อันตรกิริยาระหว่างยาของยาคลายกล้ามเนื้อที่ใช้บ่อยในทางคลินิกได้

Classification of neuropathic pain and examples of the neuroanatomical distribution of pain and sensory abnormalities

Peripheral neuropathic pain



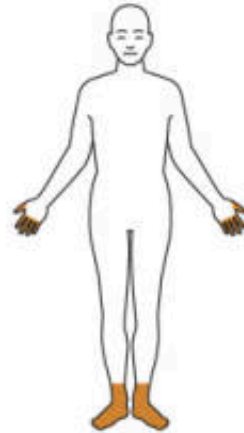
Postamputation pain
(stump and phantom pain)



Trigeminal neuralgia



Painful radiculopathy



Painful polyneuropathy

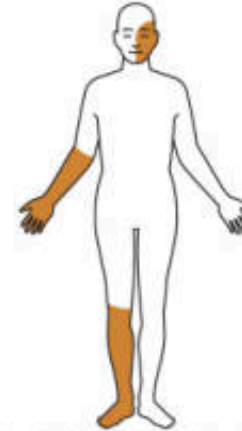


Postherpetic neuralgia

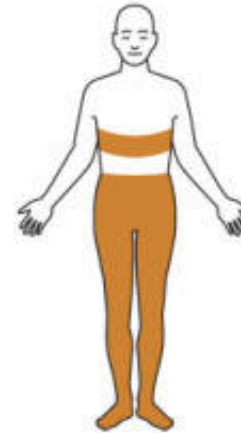


Peripheral nerve injury pain

Central neuropathic pain



Central post-stroke pain



Spinal cord injury
neuropathic pain
(at- and below level pain)



Central pain in
multiple sclerosis

- Chemotherapy-induced neuropathy
- Painful diabetic neuropathy

Symptoms

★ Negative symptoms

- ☾ Numbness, hypoalgesia

★ Positive symptoms

- ☾ Spontaneous

 - ★ Continuous: burning, tingling

 - ★ Paroxysmal: sharp shooting, lancinating, electrical shock-like

 - ★ Abnormal sensation: paresthesia, dysesthesia

- ☾ Stimulus-dependent

 - ★ Hyperalgesia

 - ★ Allodynia

Pathophysiology: Maladaptive plasticity of the somatosensory system

Peripheral sensitization	Central sensitization
• Sensitization of TRPV1, TRPM8, TRPA1 • Upregulation of sodium channel subtypes • Upregulation of $\alpha_2\delta_1$ N-type calcium channel • Proinflammatory cytokines (TNF-α, IL-1β and IL-6) and nerve growth factor (NGF) released from immune cells surrounding the nerve	• Loss of central inhibition at spinal cord ○ Dysregulation of descending inhibitory pathways (NE, 5-HT) ○ ↓ opioid binding site ○ ↓ GABA- inhibitory efficiency • Glial cell activation: release of various inflammatory mediators which facilitate spinal pain transmission

TRP = transient receptor potential ion channels

Characteristics of neuropathic pain

- ✱ Distinct pain characteristics
- ✱ Pain may persist beyond the pathology
- ✱ *Not responsive to NSAIDs/coxib*
- ✱ *Less responsive to morphine*

Drug for neuropathic pain

Antiepileptics

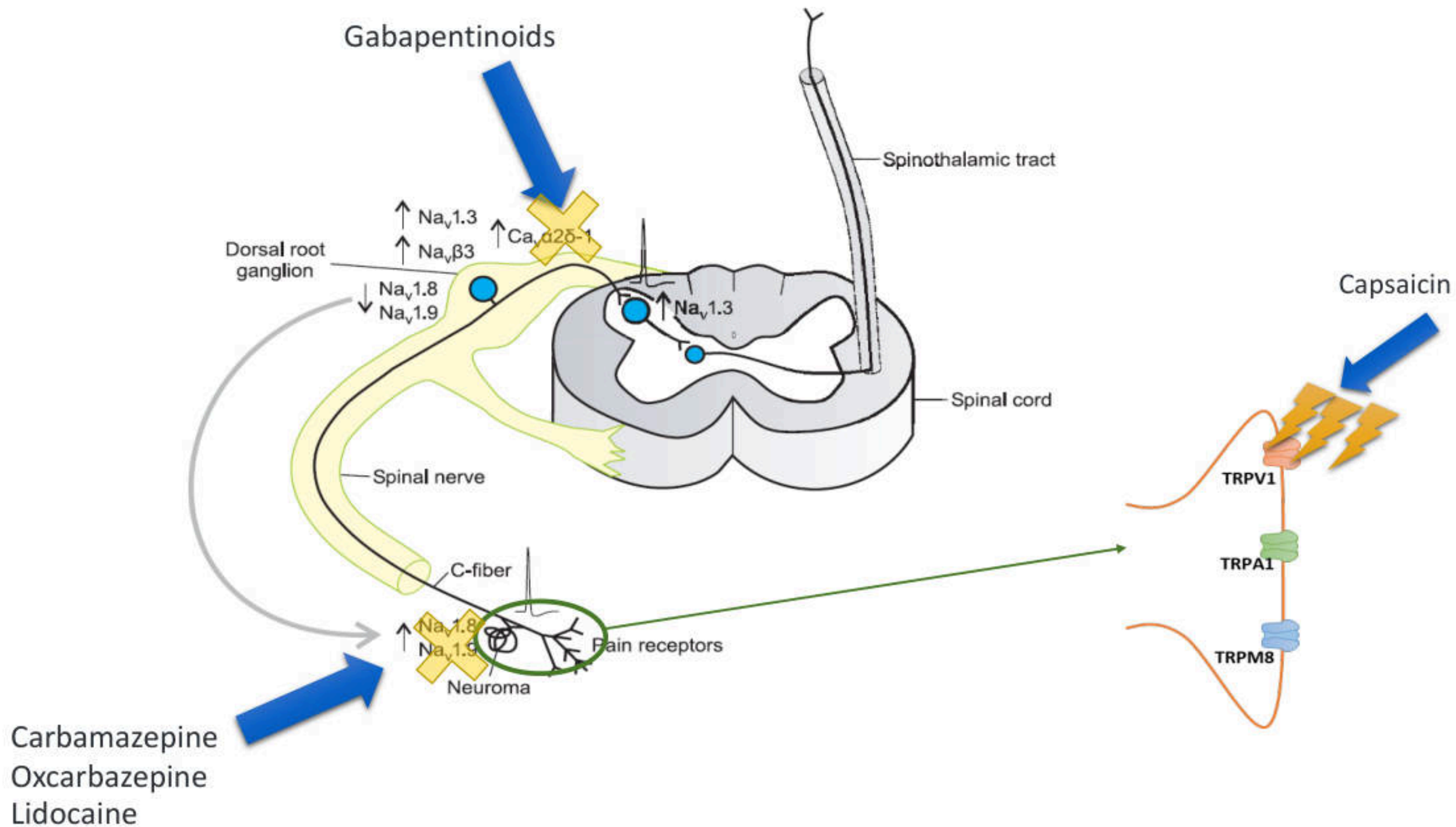
- ◆ Block voltage-gated sodium channel: carbamazepine, oxcarbazepine
- ◆ Block voltage-gated calcium channel: gabapentin, pregabalin, mirogabalin

Antidepressants

- ◆ Block 5-HT and NE reuptake transporters
- ◆ Tricyclic antidepressants: amitriptyline, nortriptyline
- ◆ Serotonin norepinephrine reuptake inhibitors: duloxetine, venlafaxine, milnacipran

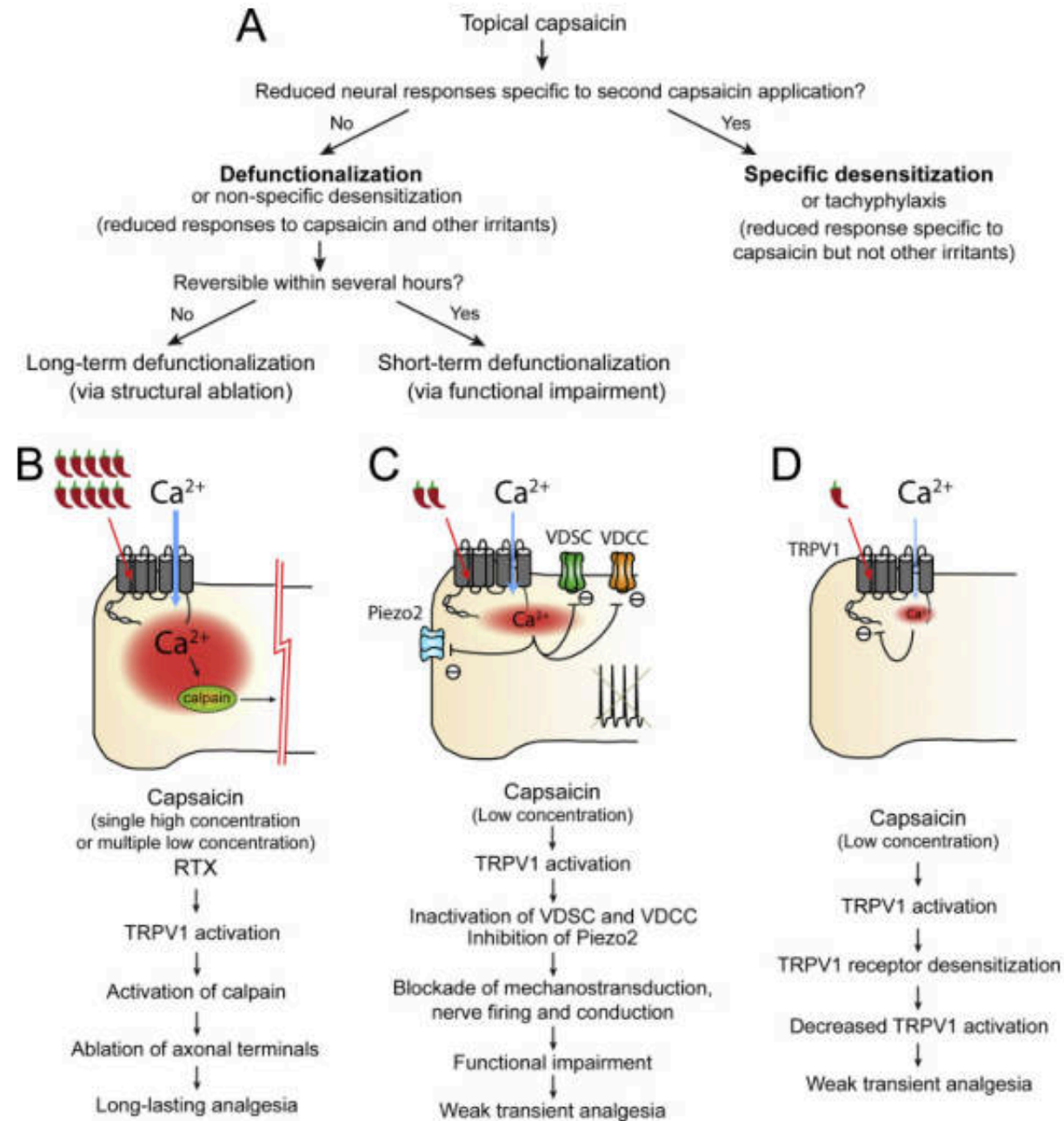
Others

- ◆ Topical capsaicin
- ◆ Ketamine
- ◆ Cannabinoids



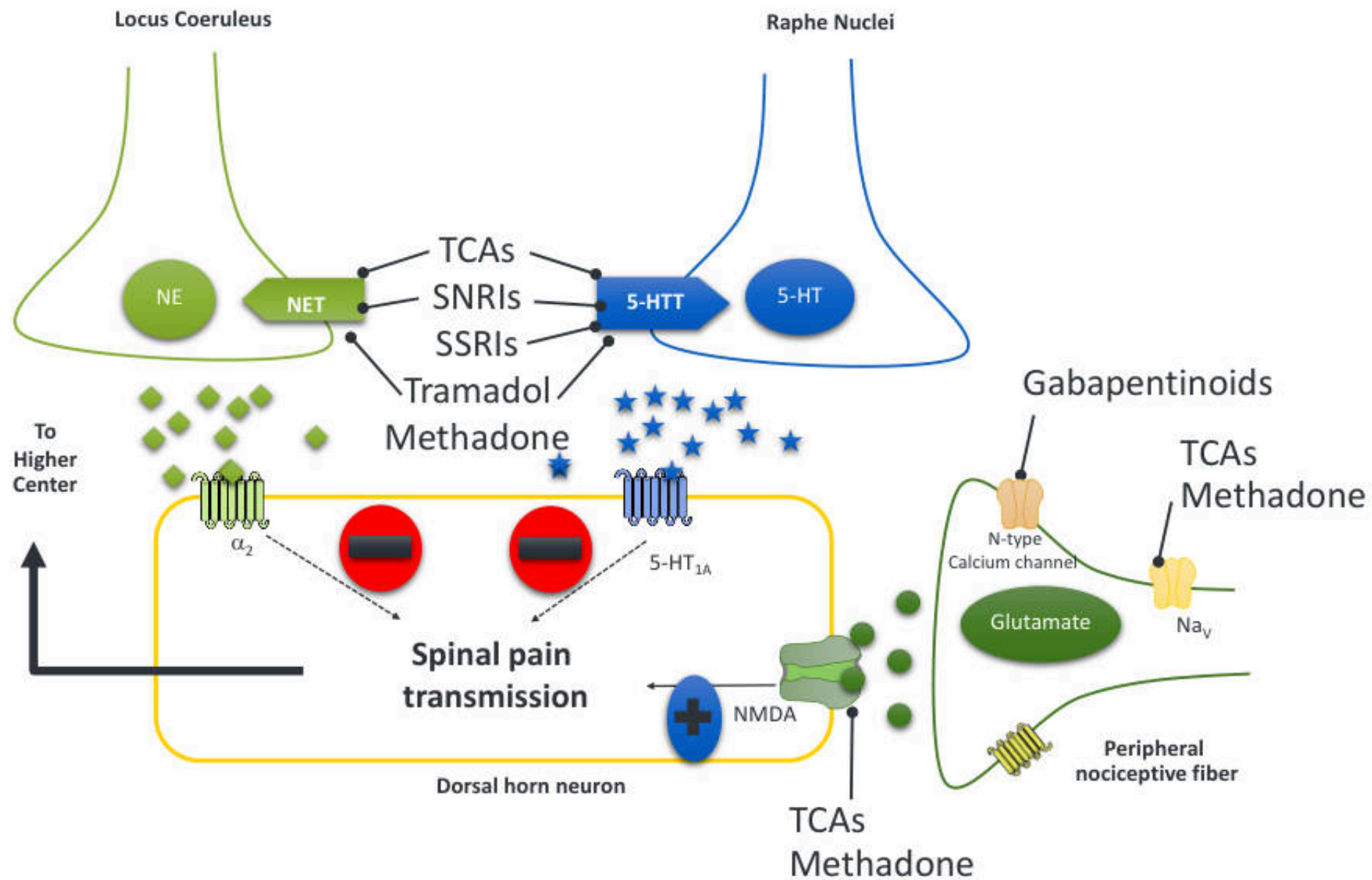
Topical capsaicin

- Activates and desensitizes TRPV1
- Initial burning sensation upon topical application
- Analgesic onset 1-2 weeks
- Effective concentration for neuropathic pain is at least 0.075%



Carbamazepine and oxcarbazepine in neuropathic pain

- ★ Accumulation/sensitization of $\text{Na}_v1.3, 1.7, 1.8, 1.9$ can lead to neuropathic pain symptoms, especially *paroxysmal sharp shooting pain*
- ★ Antiepileptics exert the analgesic effect by blockade of these Na_v at the peripheral nociceptive fibers
- ★ However, most antiepileptics that block Na_v are non-selective



Drugs used in neuropathic pain	Antineuropathic pain mechanisms	Adverse effects
$\alpha_2\delta$ -1 ligands or gabapentinoids Gabapentin, pregabalin, mirogabalin	Bind to $\alpha_2\delta$ -1 subunit of N-type voltage-gated calcium channels and inhibit glutamate release from C-fiber and neuronal hyperexcitability	Edema, weight gain, somnolence, dizziness, ataxia, incoordination
Antidepressants - TCAs: amitriptyline, nortriptyline - SNRIs: duloxetine, venlafaxine	- Block NET and 5-HTT and enhance descending inhibitory pain pathway - TCAs block NMDA, Na_v	- TCAs: Anticholinergic effect, sedation, QT interval prolongation - SNRIs: nausea, vomiting, insomnia, tachycardia, increased BP
Sodium channel blockers - Carbamazepine, oxcarbazepine - Lidocaine (transdermal patch)	Block $\text{Na}_v1.3$, $\text{Na}_v1.7$, $\text{Na}_v1.8$, $\text{Na}_v1.9$ at peripheral nociceptive fiber	- Somnolence, dizziness, ataxia, rash - Arrhythmia
Opioids - Tramadol - Methadone	- Tramadol and methadone block NET and 5-HTT - All are μ agonists - Methadone blocks NMDA	- Tramadol: nausea, vomiting - Methadone: QT interval prolongation
Capsaicin ($\geq 0.075\%$ concentration topical preparations)	Desensitize of TRPV1 and temporarily degenerate intraepidermal C- fibers	Burning sensation, irritation
Ketamine	Block NMDA receptors	Dissociative psychosis

Classification of muscle relaxants

☀ Centrally acting muscle relaxants

☾ Orphenadrine

✱ Norflex[®] orphenadrine citrate 100 mg

✱ Norgesic[®] orphenadrine citrate 35 mg, paracetamol 450 mg

☾ Tolperisone (Mydocalm[®]), Eperisone (Myonal[®])

☾ Diazepam (Valium[®]), Clonazepam (Rivotril[®])

☾ Baclofen (Lioresal[®])

☾ Tizanidine (Sirdalud[®])

☾ Riluzole (Rilutek[®]) approved for spasm in amyotrophic lateral sclerosis (ALS)



☀ Peripherally acting muscle relaxants

☾ Depolarizing muscle relaxants: succinylcholine

☾ Non-depolarizing muscle relaxants: curare alkaloid derivatives

☾ Direct-acting muscle relaxants: dantrolene



CENTRAL

Cerebral
infarction Spinal
cord
tumor

MUSCLE
SPASTICITY

clonus; stiffness; rigidity
and tightness; and
difficulty with walking,
movement, and,
occasionally, talking

Chronic
disability

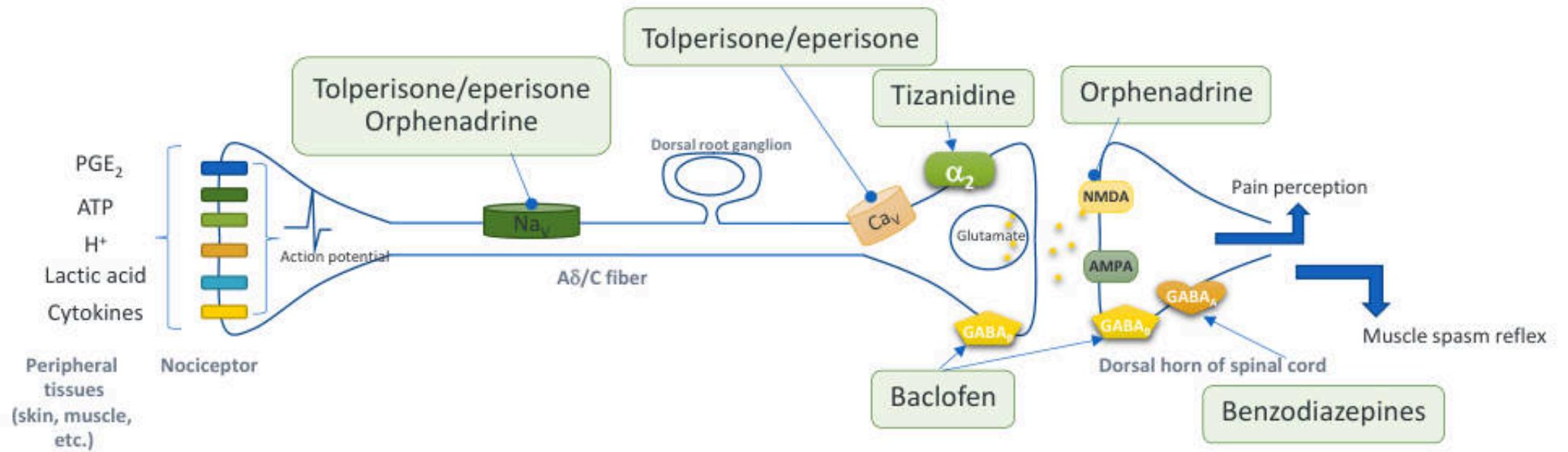
PERIPHERAL

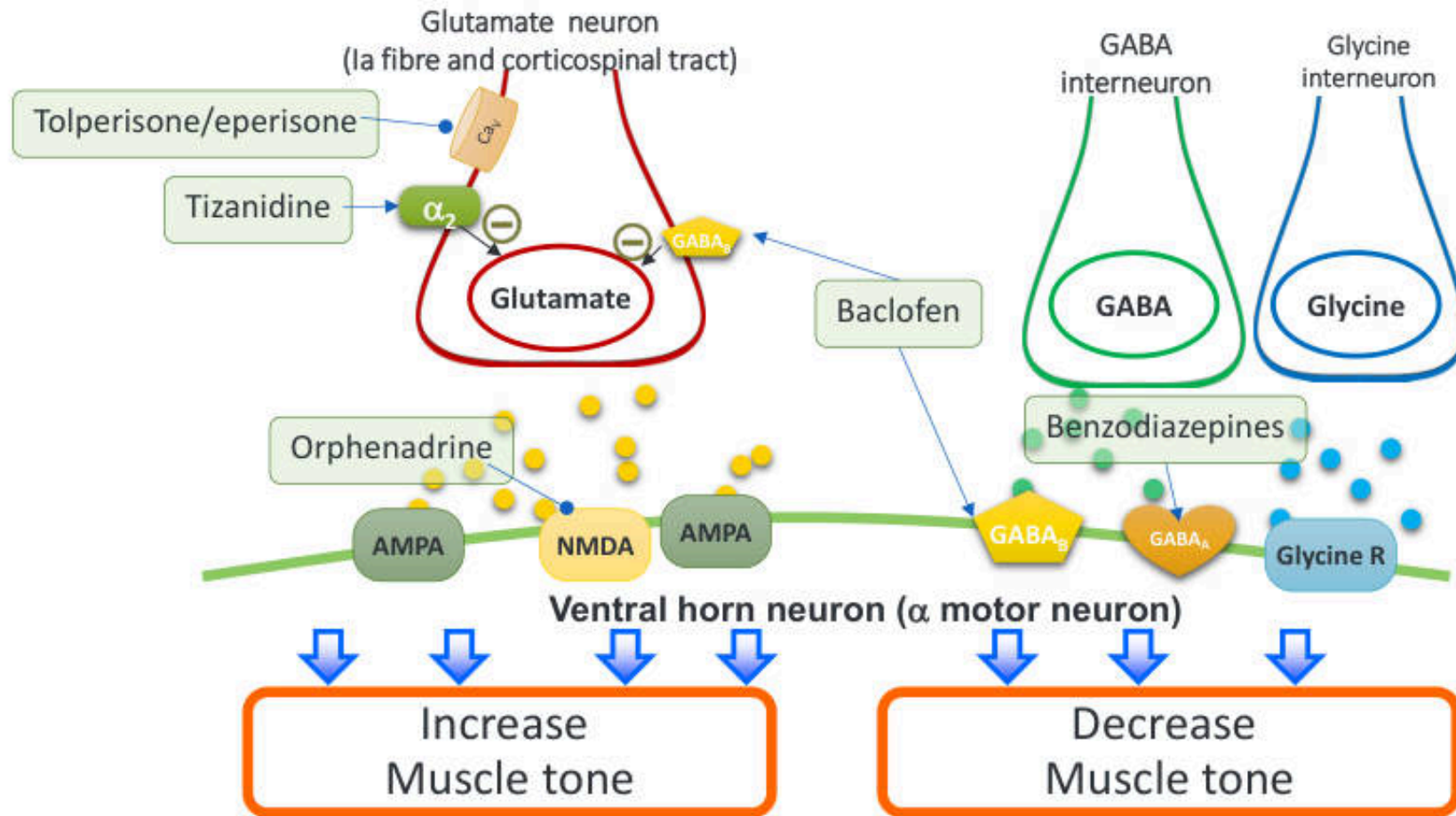
Musculoskeletal disturbances

MUSCLE
SPASM

Pain and/or
disability

Drug	Mechanism
Orphenadrine	Non-selective Na _v blockers NMDA receptor blocker (moderate affinity) Muscarinic receptor antagonist
Tolperisone Eperisone	Non-selective Na _v blockers and inhibit Ia fiber discharge Ca _v blockers and reduce neurotransmitter release from C-fiber Increase blood flow to muscle (eperisone)
Diazepam Clonazepam	GABA _A allosteric potentiating ligand
Baclofen	GABA _B agonist
Tizanidine	α ₂ agonist
Riluzole	Non-selective Na _v blockers NMDA receptor blocker





Actions at CNS, ventral horn,
motor nerve and skeletal muscle

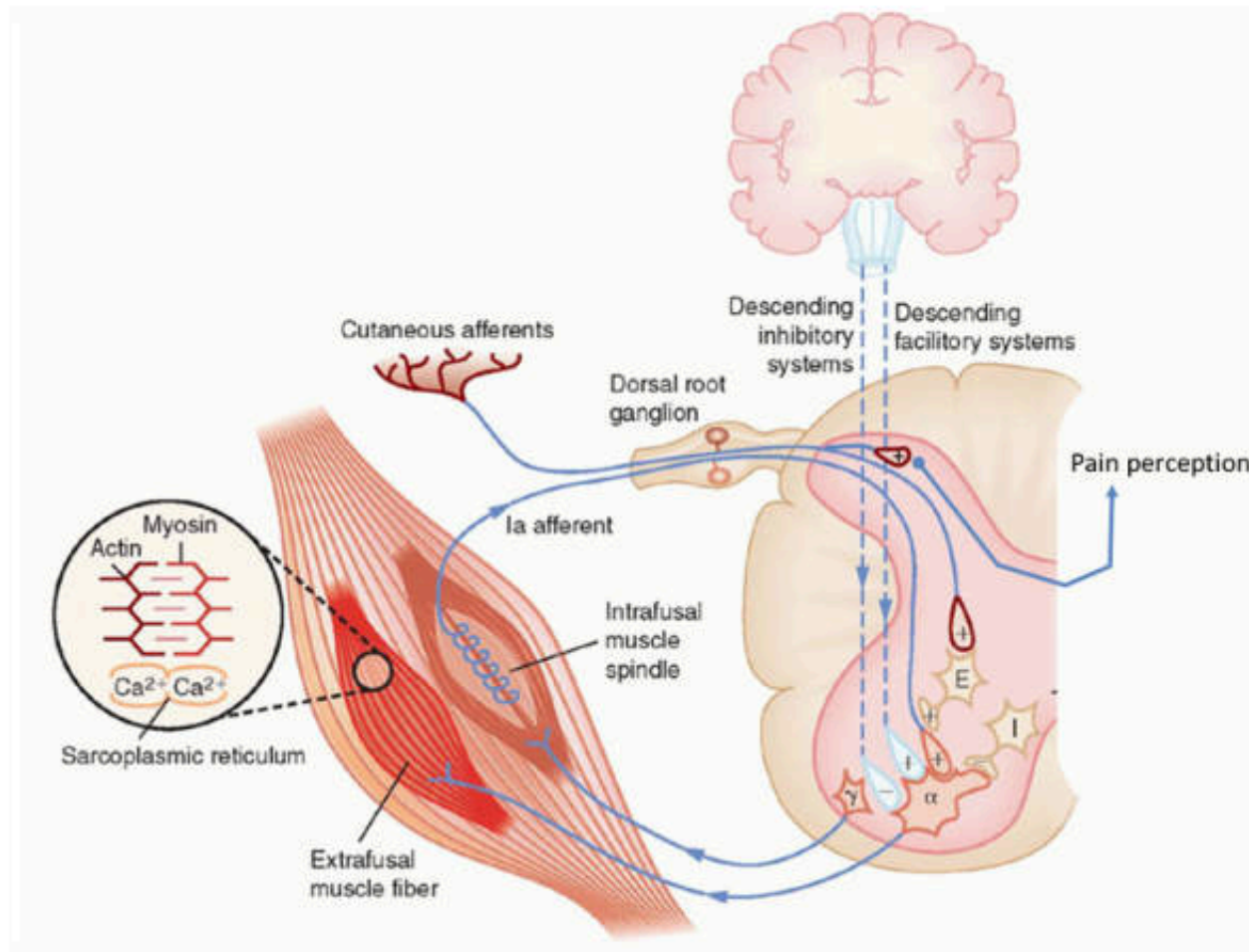


Muscle relaxant effect

Actions at sensory nerve,
dorsal horn



Analgesic effect



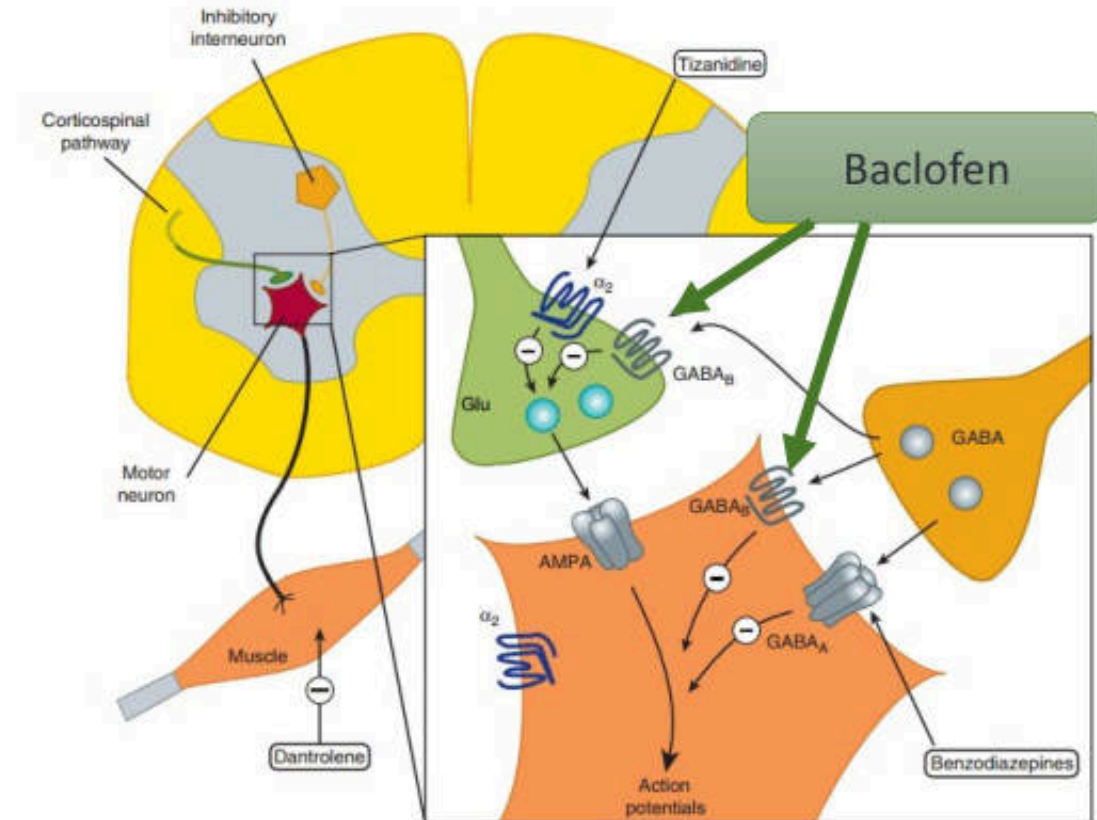
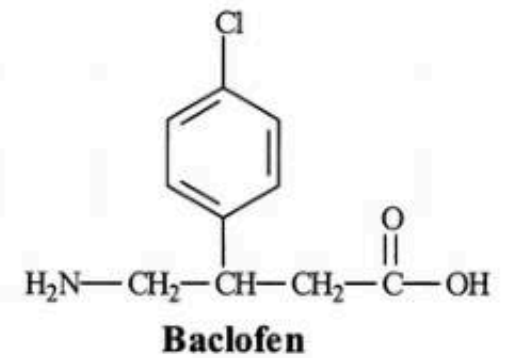
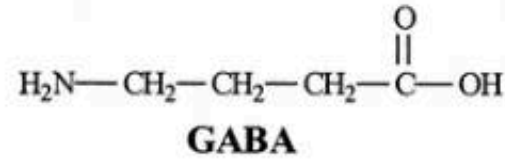
Muscle relaxants do not have
anti-inflammatory effect!!

Orphenadrine

- ✱ Produces both analgesic and anticholinergic effects
- ✱ May be dosed orally at 100 mg twice daily
- ✱ Can cause cardiovascular adverse effects such as palpitations and tachycardia so it should be used with caution in patients with heart failure, tachycardia or arrhythmias
- ✱ It also commonly causes anticholinergic side effects such as constipation, urinary retention and confusion
- ✱ Discontinuation of orphenadrine requires a taper if use becomes chronic

Baclofen

- The chemical structure resembles GABA
- Binds to GABA_B receptors to decrease synaptic transmission in the spinal cord
- Potent CNS depressing effect
- Pharmacokinetics
 - Short half-life 2-6 hr
 - Excreted mainly by kidney unchanged (70-85%)

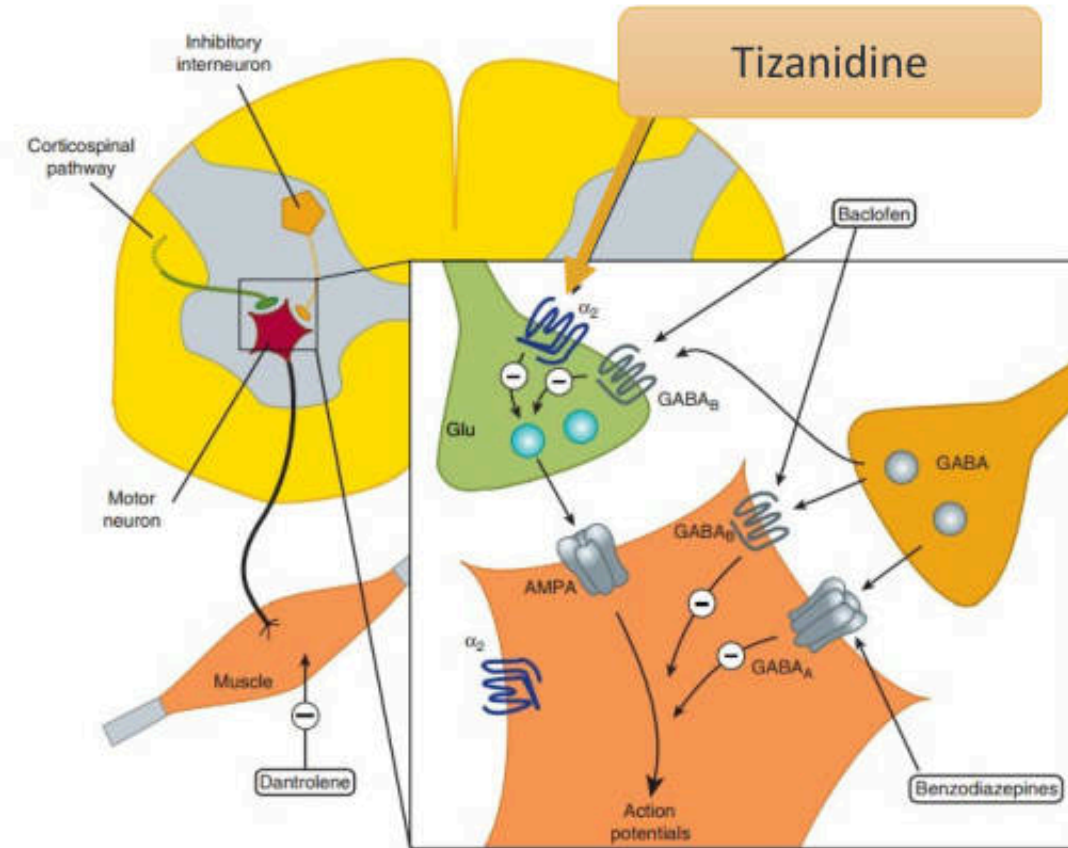


Safety use of baclofen

- ✱ Orally starting at doses of 5 mg three times a day and can be titrated up to doses of 80 mg per day given in three to four divided doses for adults
- ✱ The most common adverse effects associated with baclofen include weakness, dizziness, drowsiness, vertigo, hypotension and insomnia
- ✱ Monitor renal function
- ✱ Sudden discontinuation can result in high fever, confusion, hallucinations, seizures, and exacerbation of muscle rigidity. Caution should be taken to taper off baclofen often over several weeks.

Tizanidine

- A centrally acting α_2 adrenergic agonist
- Increase presynaptic inhibition of motor neurons blocking glutamate release
- Adverse reactions such as hypotension, bradycardia, or excessive sedation
- Require gradual dose reduction to stop therapy



Pharmacokinetics of tizanidine

- ✱ Short half-life 2-3 hr
- ✱ Metabolized extensively in the liver by CYP1A2 to inactive metabolites
- ✱ Coadministration of tizanidine with cytochrome P450 enzyme 1A2 (CYP1A2) inhibitors (fluvoxamine, ciprofloxacin) is contraindicated

Clinically relevant PK properties of muscle relaxants

Drug	CYP450 metabolism				T1/2 (Hr)	Other PK properties
	1A2	2C19	2D6	3A4		
Orphenadrine (Norflex [®] , Norgescic [®])*			?	✓	14-20	
Tolperisone (Mydocalm [®])			✓		1.5	
Eperisone (Myonal [®])				✓	1.5-2	
Diazepam (Valium [®])		✓		✓	50-100	
Clonazepam (Rivotril [®])				✓	19-60	
Baclofen (Lioresal [®])					3-4	Excreted unchanged mainly by kidney
Tizanidine (Sirdalud [®])	✓				2-3	
Riluzole (Rilutek [®])	✓				12	High fat meal reduces drug absorption

*possibly metabolised by other CYPs

Important drug interactions

PDDI

- All muscle relaxants + CNS depressants
- Orphenadrine + cholinergic drugs
- Tizanidine + drugs that cause vasodilation
 - ◆ Drugs with α_1 blocking effect
 - ◆ CCBs and other vasodilators

PKDI

- CYP1A2 inhibitors/inducers + tizanidine
 - ◆ Inhibitors: ciprofloxacin, fluvoxamine
 - ◆ Inducers: smoking, omeprazole, lansoprazole